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Answer **all** questions.

1. (a) Crude oil can be separated into simpler mixtures called fractions. The table shows information about some of the main fractions obtained from crude oil by fractional distillation.

Fraction	Boiling point range (°C)	Number of carbon atoms present in the alkanes
petroleum gases	< 20	C_1–C_4
petrol	30–75	C_5–C_8
naphtha	70–180	C_9–C_{12}
kerosene	180–250	C_{13}–C_{16}
diesel oil	250–340	C_{17}–C_{20}
lubricating oil	340–500	C_{20}–C_{24}
fuel oil	490–580	C_{25}–C_{28}

(i) Decane has a boiling point of 174 °C. Give the name of the fraction which contains decane. [1]

....

(ii) One alkane is found in both diesel oil and lubricating oil. Give the number of carbon atoms in this alkane. [1]

....

3410UD01
03

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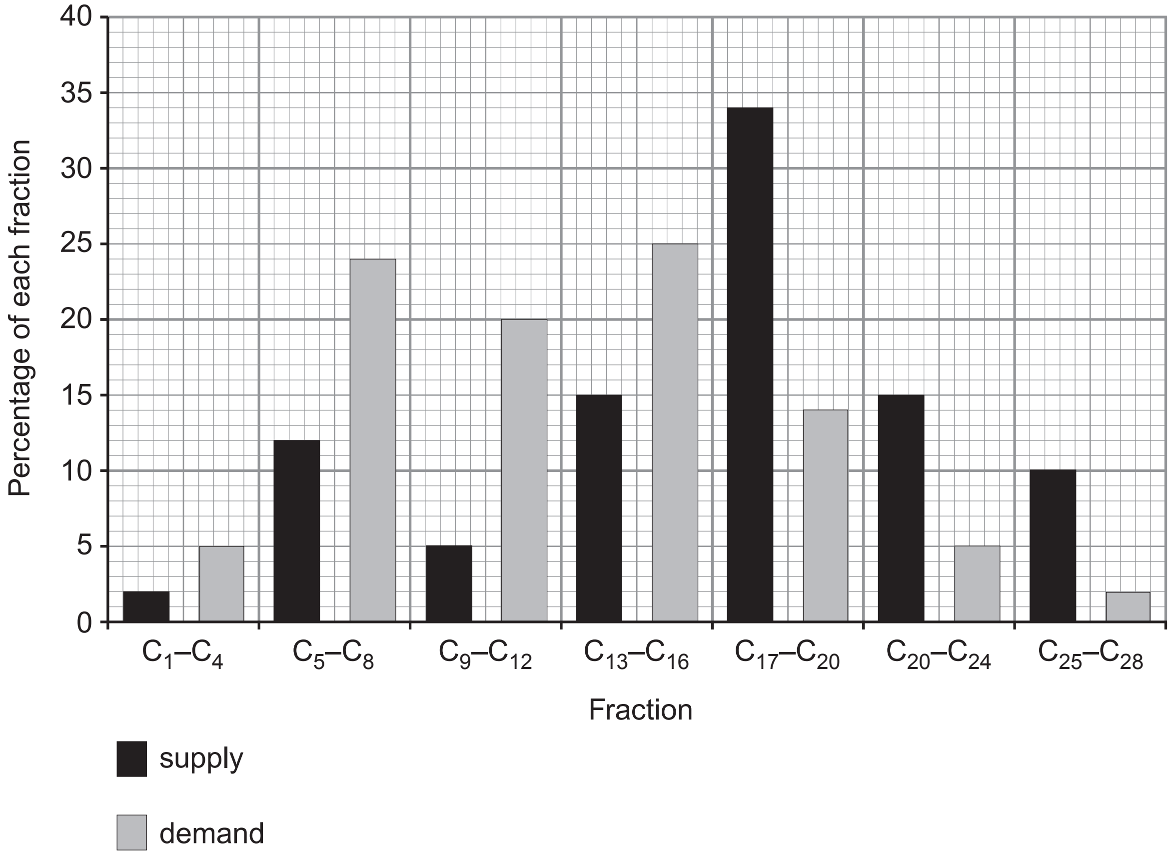
(b) The bar chart shows the supply and demand for some of the fractions obtained from crude oil.



(i) Give the fraction where the demand is 100% greater than the supply. [1]

....

(ii) Put a **tick** (✓) in the box next to the statement that best describes how the supply and demand of the fractions change as the chain length increases. [1]

- supply is greater than demand for all fractions ☐
- supply is greater than demand up to C_{16} after which demand is greater than supply ☐
- demand is greater than supply up to C_{16} after which supply is greater than demand ☐
- the difference between supply and demand increases up to C_{16} after which it decreases ☐

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(iii) Oil companies have solved the problem of over-supply of some fractions using a process called cracking.

The alkane $C_{20}H_{42}$ can be cracked forming propene, nonene and **one molecule** of hydrocarbon **A**.

I. Complete the equation for the cracking of $C_{20}H_{42}$. [1]

$$C_{20}H_{42} \longrightarrow \underset{\text{propene}}{C_3H_6} + \underset{\text{nonene}}{C_9H_{18}} + \underset{\text{hydrocarbon A}}{\dots\dots\dots\dots}$$

II. Propene is an important raw material in the production of polypropene.

Give the name of the process used to make polypropene from propene. [1]

....

3410UD01 05

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(c) Every year thousands of acres of forests are destroyed by wildfires. Fire-fighters use several different methods to put out this type of fire.

State **three** methods that are used to put out forest fires.

Give the part of the fire triangle being removed in each method.

Each method should refer to a different part of the fire triangle. [3]

Method 1 ....

Part of the fire triangle being removed ....

Method 2 ....

Part of the fire triangle being removed ....

Method 3 ....

Part of the fire triangle being removed ....

9